

SIMATS ENGINEERING

ASSESSMENT TOOL 1-Pseudocode Writing Task

COURSE NAME: R PROGRAMMING

COURSE CODE: ITA04

COURSE OUTCOME COVERED

CO: Apply R programming fundamentals including data structures, vectors, matrices, arrays, and class types to perform mathematical operations and manage data effectively.
(Bloom Level -BL2- BL4)

Assessment Tool Weightage: 20%

QUESTIONS:

Total Marks: 100

1. Variables and Data Types Analysis (BL4 – Analyze)

Design pseudocode to simulate basic variable handling in R programming:

- Declare variables of different data types (numeric, character, logical)
- Perform arithmetic and logical operations
- Display the results

2. Data Structures Implementation (BL4 – Analyze)

Write pseudocode to demonstrate advanced data structures in R:

- Create and initialize a vector, list, and matrix
- Perform operations such as indexing, updating, and aggregation
- Compare the structures based on usage

3. Matrix and Array Operations (BL4 – Analyze)

Design pseudocode to perform operations on matrices and arrays:

- Create a matrix and a 3D array
- Perform addition, multiplication (matrix), and element-wise operations
- Display the results and analyze differences

4. Data Frame Construction and Manipulation (BL4 – Analyze))

A dataset needs to be organized efficiently.

Write pseudocode to:

- Create a data frame with multiple columns (numeric and character)
- Add, delete, and modify columns
- Perform basic summary operations

5. Functions and Logical Operations (BL4 – Analyze)

Design pseudocode to implement functions and logical vector operations:

- Create a user-defined function with input parameters
- Perform operations on logical vectors
- Return and display computed results

Code of Conduct:

I Savitha R (192421368) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: SAVITHA R

1. Variables and Data Types Analysis (BL4 – Analyze)

- Declare variables of different data types (numeric, character, logical)
- Perform arithmetic and logical operations
- Display the results

SOLUTION

```
# Variables and Data Types

student_name <- "Yashwitha"
age <- 20
cgpa <- 9.2
isPresent <- TRUE

# Arithmetic Operations
total <- age + 5
square <- age^2

# Logical Operations
eligible <- age >= 18
attendance <- isPresent & eligible

# Display Results
cat("Student Name:", student_name, "\n")
cat("Age:", age, "\n")
cat("CGPA:", cgpa, "\n")
cat("Present:", isPresent, "\n")
cat("Age after 5 years:", total, "\n")
cat("Square of Age:", square, "\n")
cat("Eligible:", eligible, "\n")
cat("Attendance Status:", attendance, "\n")
```

Output:

```
> # Variables and Data Types
>
> student_name <- "Yashwitha"
> age <- 20
> cgpa <- 9.2
> isPresent <- TRUE
>
> # Arithmetic Operations
> total <- age + 5
> square <- age^2
>
> # Logical Operations
> eligible <- age >= 18
> attendance <- isPresent & eligible
>
> # Display Results
> cat("Student Name:", student_name, "\n")
Student Name: Yashwitha
> cat("Age:", age, "\n")
Age: 20
> cat("CGPA:", cgpa, "\n")
CGPA: 9.2
> cat("Present:", isPresent, "\n")
Present: TRUE
> cat("Age after 5 years:", total, "\n")
Age after 5 years: 25
> cat("Square of Age:", square, "\n")
Square of Age: 400
> cat("Eligible:", eligible, "\n")
Eligible: TRUE
> cat("Attendance Status:", attendance, "\n")
Attendance Status: TRUE
```

Values	
age	20
attendance	TRUE
cgpa	9.2
eligible	TRUE
isPresent	TRUE
square	400
student_name	"Yashwitha"
total	25

2. Data Structures Implementation (BL4 – Analyze)

Write pseudocode to demonstrate advanced data structures in R:

- Create and initialize a vector, list, and matrix
- Perform operations such as indexing, updating, and aggregation
- Compare the structures based on usage

SOLUTION :

```
# Vector
marks <- c(80, 85, 90, 88, 95)

# List
student_info <- list(
  Name = "Yashwitha",
  RollNo = 101,
  Department = "Bioinformatics",
  CGPA = 9.2
)

# Matrix
scores <- matrix(c(70,75,80,85,90,95), nrow=3)

# Access Elements
cat("First Mark:", marks[1], "\n")

# Update Vector
marks[5] <- 98

# Aggregation
cat("Highest Mark:", max(marks), "\n")
cat("Lowest Mark:", min(marks), "\n")
cat("Average Mark:", mean(marks), "\n")

# Display Structures
print(marks)
print(student_info)
print(scores)
```

OUTPUT:

```
> # Matrix
> scores <- matrix(c(70,75,80,85,90,95), nrow=3)
>
> # Access Elements
> cat("First Mark:", marks[1], "\n")
First Mark: 80
>
> # Update Vector
> marks[5] <- 98
>
> # Aggregation
> cat("Highest Mark:", max(marks), "\n")
Highest Mark: 98
> cat("Lowest Mark:", min(marks), "\n")
Lowest Mark: 80
> cat("Average Mark:", mean(marks), "\n")
Average Mark: 88.2
>
> # Display Structures
> print(marks)
[1] 80 85 90 88 98
> print(student_info)
$Name
[1] "Yashwitha"

$RollNo
[1] 101

$Department
[1] "Bioinformatics"

$CGPA
[1] 9.2
> print(scores)
     [,1] [,2]
[1,]   70   85
[2,]   75   90
[3,]   80   95
```

Data

scores	num [1:3, 1:2] 70 75 80 85 90 95	
student_info	List of 4	

Values

marks	num [1:5] 80 85 90 88 98
-------	--------------------------

3. Matrix and Array Operations (BL4 – Analyze)

Design pseudocode to perform operations on matrices and arrays:

- Create a matrix and a 3D array
- Perform addition, multiplication (matrix), and element-wise operations
- Display the results and analyze differences

SOLUTION:

```
# Create Matrices
A <- matrix(c(2,4,6,8), nrow=2)
B <- matrix(c(1,3,5,7), nrow=2)

cat("Matrix A\n")
print(A)

cat("Matrix B\n")
print(B)

# Addition
result1 <- A + B

# Matrix Multiplication
result2 <- A %% B

# Element-wise Multiplication
result3 <- A * B

cat("Matrix Addition\n")
print(result1)

cat("Matrix Multiplication\n")
print(result2)

cat("Element-wise Multiplication\n")
print(result3)

# Create 3D Array
array_data <- array(10:33, dim=c(2,3,4))

cat("3D Array\n")
print(array_data)
```

OUTPUT:

```
>
> cat("Element-wise Multiplication\n")
Element-wise Multiplication
> print(result3)
      [,1] [,2]
[1,]    2   30
[2,]   12   56
>
> # Create 3D Array
> array_data <- array(10:33, dim=c(2,3,4))
>
> cat("3D Array\n")
3D Array
> print(array_data)
, , 1
      [,1] [,2] [,3]
[1,]   10   12   14
[2,]   11   13   15
, , 2
      [,1] [,2] [,3]
[1,]   16   18   20
[2,]   17   19   21
, , 3
      [,1] [,2] [,3]
[1,]   22   24   26
[2,]   23   25   27
, , 4
      [,1] [,2] [,3]
[1,]   28   30   32
[2,]   29   31   33
```

Data		
A	num [1:2, 1:2] 2 4 6 8	
B	num [1:2, 1:2] 1 3 5 7	
result1	num [1:2, 1:2] 3 7 11 15	
result2	num [1:2, 1:2] 20 28 52 76	
result3	num [1:2, 1:2] 2 12 30 56	
Values		
array_data	int [1:2, 1:3, 1:4] 10 11 12 13 14 15 ...	

4. Data Frame Construction and Manipulation (BL4 – Analyze)

A dataset needs to be organized efficiently.

- Create a data frame with multiple columns (numeric and character)
- Add, delete, and modify columns
- Perform basic summary operations

SOLUTION:

```
# Create Data Frame
student <- data.frame(
  Name = c("Yashwitha", "Rahul", "Priya"),
  Age = c(20, 21, 20),
  Marks = c(91, 87, 89)
)

cat("Original Data Frame\n")
print(student)

# Add New Column
student$Result <- c("Pass", "Pass", "Pass")

# Modify Marks
student$Marks[1] <- 95

# Delete Age Column
student$Age <- NULL

cat("Modified Data Frame\n")
print(student)

cat("Summary\n")
summary(student)
```

OUTPUT:

```
Original Data Frame
> print(student)
  Name Age Marks
1 Yashwitha 20 91
2 Rahul 21 87
3 Priya 20 89
>
> # Add New Column
> student$Result <- c("Pass", "Pass", "Pass")
>
> # Modify Marks
> student$Marks[1] <- 95
>
> # Delete Age Column
> student$Age <- NULL
>
> cat("Modified Data Frame\n")
Modified Data Frame
> print(student)
  Name Marks Result
1 Yashwitha 95 Pass
2 Rahul 87 Pass
3 Priya 89 Pass
>
> cat("Summary\n")
Summary
> summary(student)
      Name           Marks           Result
Length:3      Min.   :87.00  Length:3
Class :character 1st Qu.:88.00  Class :character
Mode  :character Median :89.00  Mode  :character
              Mean   :90.33
              3rd Qu.:92.00
              Max.   :95.00
```

Data

student

3 obs. of 3 variables



5. Functions and Logical Operations (BL4 – Analyze)

Design pseudocode to implement functions and logical vector operations:

- Create a user-defined function with input parameters
- Perform operations on logical vectors
- Return and display computed results

SOLUTION:

```
# User-defined Function
studentDetails <- function(name, marks)
{
  cat("Student Name:", name, "\n")
  cat("Marks:", marks, "\n")

  if(marks >= 50)
  {
    cat("Result: Pass\n")
  }
  else
  {
    cat("Result: Fail\n")
  }
}

# Function Call
studentDetails("Yashwitha", 91)

# Logical Vector
attendance <- c(TRUE, TRUE, FALSE, TRUE, FALSE)

cat("Attendance Vector\n")
print(attendance)

cat("Number of Present Days:", sum(attendance), "\n")

cat("NOT Attendance\n")
print(!attendance)
```

OUTPUT:

```
Max.      :95.00
> # User-defined Function
> studentDetails <- function(name, marks)
+ {
+   cat("Student Name:", name, "\n")
+   cat("Marks:", marks, "\n")
+   if(marks >= 50)
+   {
+     cat("Result: Pass\n")
+   }
+   else
+   {
+     cat("Result: Fail\n")
+   }
+ }
>
> # Function Call
> studentDetails("Yashwitha", 91)
Student Name: Yashwitha
Marks: 91
Result: Pass
>
> # Logical Vector
> attendance <- c(TRUE, TRUE, FALSE, TRUE, FALSE)
>
> cat("Attendance Vector\n")
Attendance Vector
> print(attendance)
[1] TRUE TRUE FALSE TRUE FALSE
>
> cat("Number of Present Days:", sum(attendance), "\n")
Number of Present Days: 3
>
> cat("NOT Attendance\n")
NOT Attendance
> print(!attendance)
[1] FALSE FALSE TRUE FALSE TRUE
```

R ▾

 Global Environment ▾

Values

attendance	logi [1:5] TRUE TRUE FALSE TRUE FALSE
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Functions

studentDetails	function (name, marks)	
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